

Hull Tactical Market Prediction

Tactical Asset Allocation with Machine Learning

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The Kaggle Challenge

- **Goal:** Predict daily S&P 500 excess returns and convert to weights in $[0, 2]$
- **Metric:** Adjusted Sharpe ratio (penalizes risk and underperformance)
- **Data:** Anonymized macroeconomic, sentiment, and pricing features ($\sim 9k$ days)
- **Constraint:** Weights must be between 0 and 2; predictions evaluated daily

Why this matters

Tactical asset allocation dynamically adjusts market exposure based on predictive signals. The adjusted Sharpe's penalties for risk and underperformance make it a highly realistic performance measure.

Adjusted Sharpe Ratio

$$\text{adjusted_sharpe} = \frac{\text{Sharpe}}{\text{vol_penalty} \times \text{return_penalty}}$$

$$\text{vol_penalty} = 1 + \max\left(0, \frac{\sigma_{\text{strat}}}{\sigma_{\text{mkt}}} - 1.2\right)$$

$$\text{return_penalty} = 1 + \frac{\text{gap}^2}{100}$$

$$\text{gap} = \max(0, (\mu_{\text{mkt}} - \mu_{\text{strat}}) \times 100 \times 252)$$

Key insight

The **quadratic** return penalty makes sustained underperformance catastrophic (falling 5% behind multiplies the denominator by ~ 1.25). The strategy **defaults to index exposure** ($w = 1.0$) unless highly confident.

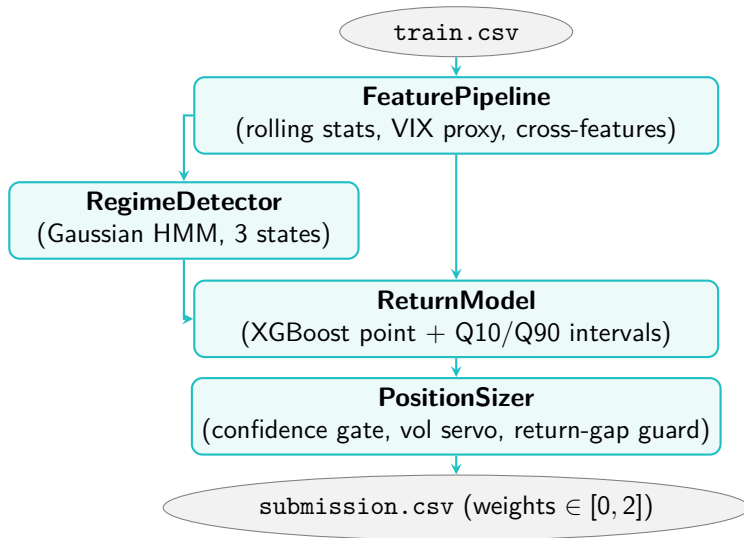
- **Train set:** 8 990 trading days with forward returns, risk-free rate, and anonymized features
 - Columns: I1–I5 (interest rates), E1–E12 (economic), S1–S5 (sentiment), P1–P11 (pricing), V1–V13 (volatility)
- **Project data:** Only `train.csv`, split chronologically (no random shuffling)

Temporal split (no random shuffling!)

Tuning block	Purge gap	Holdout
7 290 rows	200 rows	1 500 rows

The purge gap prevents autocorrelation leakage between train and evaluation data.

System Overview



Rolling Statistics

- Volatility: $\sigma_{5,20,60,252}$
- Momentum: $r_{5,20,60,252}$
- Sharpe ratios: $SR_{20,60}$
- Drawdown: DD_{60} , $MaxDD_{252}$

VIX Features

- VIX Proxy: mean of V1–V13
- Percentile rank (train-frozen)
- Term-slope: $V13 - V1$
- VIX delta (history-buffered)

Cross-Features

- Yield-curve proxy: $I_{last} - I_{first}$
- Macro-sentiment divergence
- Price-momentum agreement
- Autocorrelation: $\rho_{5,20}$

Anti-Leakage

- Median imputation: train medians only
- 400-row history buffer for rolling windows
- Lagged targets only: `.shift(1)`

Total: ~ 107 engineered features after availability filtering.

VIX Proxy Design

- The V1–V13 columns in `train.csv` are anonymized volatility features
- A **parameter-free proxy** maps V1–V13 into a unified volatility level
- Calculated as the cross-sectional mean: $\text{VIX Proxy}_t = \frac{1}{13} \sum_{i=1}^{13} V_{i,t}$

Why a proxy?

VIX is a well-known fear gauge. Compressing the volatility dimensions recovers a signal that is:

- Statistically associated with future volatility
- Consistent with market microstructure and fear-gauge literature
- Highly useful for percentile ranking and delta (momentum) features

VIX Proxy, Derived Features

Four features are derived from the VIX proxy:

- 1 **VIX proxy**, cross-sectional mean level signal
- 2 **VIX percentile rank**, where current VIX sits vs. historical distribution
- 3 **VIX term-slope**, $V_{13} - V_1$ (volatility term structure)
- 4 **VIX delta**, day-over-day change (momentum of fear)

Percentile rank anti-leakage

During `fit_transform`, the sorted VIX array is frozen from training data. During transform, test VIX values are ranked against this frozen reference using `np.searchsorted` to prevent lookahead leakage.

Market Regime Detection

- **Gaussian HMM** with 3 hidden states, diagonal covariance
- Inputs: `vix_proxy_pct`, `roll_vol_20`, `momentum_20`, `autocorr_20`

Output per row

- `regime_state`, Viterbi-decoded hidden state
- `regime_prob_{0,1,2}`, posterior probabilities
- `regime_stability`, confidence in regime assignment
- `regime_label`, semantic labels (e.g. `bull_low_vol`, `bear_high_vol`) assigned by training set mean return/volatility comparisons

Return Prediction Model (XGBoost)

Three models trained simultaneously on `market_forward_excess_returns`:

Model	Objective	Purpose
Point model	reg:squarederror	Direction & magnitude
Lower quantile (Q10)	reg:quantileerror, $\alpha = 0.1$	Lower bound
Upper quantile (Q90)	reg:quantileerror, $\alpha = 0.9$	Upper bound

Confidence signal:

$$\text{raw_conf} = \frac{|\text{point_pred}|}{\max(\text{Q90} - \text{Q10}, \varepsilon)} \longrightarrow \text{confidence} = \text{clip}\left(\frac{\text{raw_conf} - p_{10}}{p_{90} - p_{10}}, 0, 1\right)$$

Percentiles p_{10}, p_{90} are computed on training data and frozen. This maps raw confidence to $[0, 1]$, filtering noise while preserving relative ordering.

XGBoost parameters

- `max_depth` = 4
- `learning_rate` = 0.05
- `subsample` = 0.8
- `colsample_bytree` = 0.8

Tree Count & Regularization

- Point model uses **400 trees** to learn directional signal
- Quantile bounds are limited to **150 trees** ($\frac{1}{3}$ of point)
- Shorter ensembles act as early stopping to prevent overfitting to noisy tails

Purged Walk-Forward CV: Time-series cross-validation with a configurable purge gap between each training window and validation fold, preventing any forward information leakage through autocorrelated targets.

Confidence-Gated Position Sizing

Two-branch logic per time step:

```
if confidence < threshold:  w = 1.0   (index-tracking, no return-gap risk)
else:                       w = 1.0 + gain × pred × vol_scale × reg_mult
```

Volatility scaling (Moreira & Muir, 2017):

$$\text{vol_scale} = \text{clip}\left(\frac{\sigma_{\text{target}}}{\sigma_t}, 0.5, 2.0\right)$$

Leverage is dynamically scaled inversely to current market volatility to maintain stable target risk.

Regime-Based Position Multipliers

Active risk scaling adapts to the detected market state:

Regime	Multiplier
Bull, low volatility	1.30
Normal	1.00
Transition	0.70
Bear, high volatility	0.50

Rationale

Regime multipliers scale active bets up in favorable environments and scale them down in highly volatile or transition regimes to mitigate downside risk.

Two sequential guards applied after the base weight calculation:

- **Volatility servo:** If rolling strategy vol / market vol exceeds 1.149 (soft cap) or 1.20 (hard cap), active deviation is scaled back to 55% or 20% to avoid the vol penalty.
- **Return-gap guard:** If strategy return falls behind the market, force $w \geq 1.0$ to prevent further divergence and avoid the quadratic return penalty.

Defensive Mechanism

The quadratic return penalty makes underperformance catastrophic. Guards act defensively, preventing severe drawdown and underperformance without limiting upside.

Walk-Forward Backtesting

- **WalkForwardBacktester** re-fits the **entire pipeline** independently in each fold
- Features, regime, model, and strategy are all refit from scratch per fold
- No state, scaler, or model weight is shared between folds
- Each fold is scored with the exact competition metric

Fold-by-Fold Validation Diagnostics

Fold	Validation Range	Adj. Sharpe	Mean Weight	Dominant Regime
0	3 790 – 4 372	+0.9985	0.914	Transition
1	4 373 – 4 955	-0.2741	1.136	Transition
2	4 956 – 5 538	+0.8578	0.981	Bear (High Vol)
3	5 539 – 6 121	+1.5707	1.012	Transition
4	6 122 – 6 704	+0.9661	0.950	Bear (High Vol)
5	6 705 – 7 287	+0.9965	0.930	Transition

Two-Stage Tuning Strategy

- ① **Stage 1**, joint model + strategy sweep
 - Features built once; 100 Optuna trials
 - Each trial refits the full model (~90 min total)
 - Initial baseline strategy CV adjusted Sharpe: **0.7246**
 - Purpose: identify a good model parameter region
- ② **Stage 2**, strategy-only sweep
 - Features **and** model pre-built once
 - 300 trials sweep strategy params only (~50 s total)
 - Best fine-tuned CV adjusted Sharpe: **0.8526**

Why separate stages?

Jointly optimising model and strategy parameters confounds the two. The strategy parameters (gain, threshold, vol cap) interact strongly with the competition metric's specific penalty structure and benefit from a dedicated, faster search.

Holdout Results

Configuration	Adj. Sharpe	Ann. Return	Vol Ratio	Active Days
This model (Tuned)	0.8605	+20.2%	1.020	29.5%
Baseline ($w = 1.0$)	0.7151	+16.8%	1.000	—
Delta	+0.1454	+3.4 pp		

- **+20.3%** improvement in adjusted Sharpe over baseline on untouched holdout
- Inner CV mean adjusted Sharpe: **0.8526** (6 folds)
- Only 29.5% of days are active, meaning the strategy is highly selective
- Tuned parameters: confidence threshold = 0.364, gain = 179.3, target vol = 0.298, soft vol cap = 1.149

Leakage Prevention

Risk	Mitigation
Feature leakage from future	<code>fit_transform</code> on training only; apply frozen stats
Rolling boundary leakage	400-row history buffer prepended to test/holdout
Temporal lookahead in CV	Purged walk-forward CV with 200-day purge gap
Target leakage	Lagged targets only (<code>.shift(1)</code>); target excluded from features
Tuning lookahead	Hyperparameters tuned on inner temporal holdout
XGBoost non-determinism	Enforce <code>nthread=1</code> , <code>tree_method='hist'</code> , <code>seed=42</code>

Critical Importance

Strict anti-leakage controls prevent inflated backtest performance. Purge gaps and history buffering are mandatory for reliable time-series modeling.

Reproducibility measures:

- All random seeds: 42 (XGBoost, sklearn, HMM)
- XGBoost: `nthread=1`, `tree_method='hist'` for cross-platform determinism
- Log `holdout_size` and `purge_gap` with every experiment

Why only 29.5% active days?

- The confidence threshold is high, as most days the prediction interval is wide relative to the point estimate
- Active days cluster around high-confidence signals: trend-consistent predictions during low-volatility bull regimes
- Index-tracking as the default absorbs return-gap risk; the strategy can only improve the score by being active

Key Takeaways:

- The adjusted Sharpe's quadratic return penalty fundamentally shapes the strategy design.
- Confidence-gating is essential: default to the index unless high confidence is present.
- Volatility management (Moreira & Muir, 2017) boosts Sharpe ratio by scaling risk dynamically.
- HMM-based regime detection provides context to optimize position sizing.
- Purged walk-forward validation is mandatory to estimate real out-of-sample performance.

References:

- Moreira, A. & Muir, T. (2017). *Volatility-Managed Portfolios*. Journal of Finance, 72(4), 1611–1644.
- Kaggle Competition:
<https://www.kaggle.com/competitions/hull-tactical-market-prediction>
- Chen, T. & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. KDD 2016.